QR-012 Bag leaks and dry spots

Doc ID	QR-012
Title	Bag leaks and dry spots
Parent	CS-006 §7.2; CS-006 §7.4; CS-006 §7.5; CS-006 §7.7; CS-007 §7.3; CS-007 §7.5
Checked against	CS-006 Rev D; CS-006 Rev D; CS-006 Rev D; CS-006 Rev D; CS-007 Rev D; CS-007 Rev D

Derived from its parent standards. If this and a parent disagree, the parent is right.

1. My bag won't hold vacuum: where to look for the leak

Look at the tacky tape seams first, then the pleat folds, the connector bases and the spiral ends, which can pierce the bag. If the gauge will not pass 25 inHg, hunt now: press the seams, listen, re-seat pleats and connector bases (CS-006 §7.4).

Also called: leak, leaking, air leak, losing vacuum, bag won't hold, won't pull down, vacuum leak, bag leak.

1. Press along every tacky tape seam, all the way round (CS-006 §7.4 step 1).
2. Listen for a hiss.
3. Re-seat each pleat fold.
4. Re-seat each through-bag connector base.
5. Check the spiral-wrap ends: a sharp cut end pierces the bag (CS-006 §7.4 step 3). Tape them over.
6. Soapy water only on hose fittings and connectors outside the bag: brush it on and watch for bubbles. Never on bag film, pleats or tape seams over a dry stack; find those by pressing and listening (CS-006 §7.7).
7. Pull vacuum again and re-run the drop test.

Symptom	Likely cause	Do (CS-006 §7.7)
Gauge won't reach 25 inHg	Perimeter leak, connector base, pleat fold	Press seams, re-seat. Soapy water only on fittings and connectors outside the bag, never on the bag over a dry stack
Passes 27, then creeps down	Micro-leak: spiral end or bag puncture	Patch with a square of tacky tape; drop test again
Leak you cannot find	Air coming through the mold, or a sharp edge piercing the bag	See QR-012 §3

The same hunt applies to a wet layup bag (CS-007 §7.5 step 2). Do not mix resin for an infusion until the drop test passes (CS-006 §7.4 step 3).

2. What passes the drop test? (inHg numbers for infusion and wet layup)

Infusion: pull to 27 inHg or better, isolate the pump, and lose 1 inHg or less over 10 minutes, then open the valve again (CS-006 §7.4). Wet layup: pull to 25 inHg or better and lose 2 inHg or less over 10 minutes (CS-007 §7.5).

Also called: drop test, leak test, leak-down, leakdown, vacuum test, pass fail, inHg.

Process	Vacuum before the test	Pass	Where the readings go
Infusion	27 inHg or better	1 inHg or less lost in 10 min	Note on "Drop test, 1 inHg or less over 10 min"
Wet layup	25 inHg or better	2 inHg or less lost in 10 min	Note on "Wet layup and bag" ("Wet layup over core" on Foam wrapped)

Sources: CS-006 §7.4 steps 1 and 2; CS-007 §7.5 steps 1 and 2.

1. Pull to the vacuum in the table.
2. Close the valve to isolate the pump. The gauge must be on the bag side of the closed valve, never on the pump (CS-006 §5; CS-006 §7.4 step 2).
3. Read the gauge.
4. Wait 10 minutes.
5. Read it again.
6. Write both readings in the step note.
7. On a pass, open the valve again and check the gauge climbs back to the vacuum in the table before you open the inlet or walk away (CS-006 §7.4 step 4; CS-007 §7.5 step 4).

On an infusion the drop-test step will not take a member's signature without the note, and "Infuse" cannot be signed until it is (CS-006 §7.4 step 2). A lead can sign without it (Sign without it), and the reason goes in the work order's event log. Fail means keep hunting and test again; no resin until it passes (CS-006 §7.4 step 3).

3. Air leaking in from nowhere (a leak you cannot find)

If you have pressed every seam and still lose vacuum, look next at the mold itself (an unsealed back or side, or a crack), and at a sharp mold edge or mesh corner piercing the bag from inside (CS-006 §7.7).

Also called: leak you can't find, mystery leak, leak from nowhere, air through the mold, pinhole, bag puncture.

1. Sand or radius sharp mold edges, or edge them with breather (CS-006 §7.7).
2. Chamfer or tape down the flow mesh corners.
3. Check whether the mold's back and sides were sealed. Sealing them is shop practice; CS-004 does not yet require it (CS-006 §7.7).
4. Check the mold for cracks.
5. Patch any bag puncture with a square of tacky tape.
6. Drop test again.

This fix list is shop practice (Larry, #composites 2026-04-03), recorded in CS-006 §7.7.

If the gauge passes 27 inHg and then creeps down, it is more likely a micro-leak at a spiral end or a small bag puncture: patch it with a square of tacky tape and drop test again (CS-006 §7.7).

4. Dry spot or resin not flowing during an infusion (resin still wet)

A dry patch far from the inlet while the resin is still wet: call the experienced infuser about an emergency second feed, and do not pierce the bag yourself (CS-006 §7.7). The usual causes are mesh that did not cover the area, or resin travel over 0.5 m from the feed.

Also called: dry spot, dry patch, starved area, unwetted, white fibre, white fiber, resin starved, flow stopped, resin not flowing, stopped flowing, flow front stalled, stuck.

1. While the resin is still wet, call the experienced infuser (CS-006 §7.1 step 3) about an emergency second feed. Do not pierce the bag yourself (CS-006 §7.7).

2. Write what happened and where on the work order, with a photo (CS-006 §7.5 step 6).

Prevent it next time (CS-006 §7.2):

- Flow mesh over the entire part, stopping 30–50 mm short of the outlet side.
- Resin travel inlet to outlet about 0.5 m at most. A larger part needs more feeds, planned on the work order.
- Mix enough: (dry fiber mass × 0.65) × 1.2–1.3, rounded up. Starving an infusion mid-pull cannot be recovered (CS-006 §7.5 step 1).

Full wet-out is checked through the bag at clamp time and again after demould: no dry patches (CS-006 §8).

5. Dry spot found after the part has cured

A dry patch in a cured part cannot be fixed by more resin through the bag. Record it as a defect on the work order and assess it against the part's quality checks (CS-006 §7.7). Photograph it.

Also called: dry spot, dry patch, cured dry spot, defect, white fibre, white fiber, starved laminate, porosity.

1. Photograph the dry area (CS-006 §7.5 step 6).
2. Record it as a defect on the work order (CS-006 §7.7).
3. Assess it against the work order's quality checks.
4. Note for the next run what caused it: mesh coverage, travel distance over 0.5 m, or not enough resin (CS-006 §7.2; CS-006 §7.5 step 1).

The full wet-out check is no dry patches visible, at clamp time and after demould, recorded in the work order's quality checks with photos (CS-006 §8).

For a wet layup, dry patches are caught ply by ply: squeegee each ply until it is shiny-wet with the weave visible, and chase every dry, white patch before the next ply goes on, because you will not see it again (CS-007 §7.3 step 1).

6. Race-tracking: resin racing along one edge

Race-tracking is resin finding a fast path, usually along an edge, and reaching the outlet before wetting the middle. Clamp the inlet briefly, rely on the mesh break, and note it for the next mold: tape a dam (CS-006 §7.7).

Also called: race-tracking, racetracking, racing, edge flow, resin running along the edge.

1. Clamp the inlet briefly (CS-006 §7.7).
2. Open the inlet again slowly, as at the start (CS-006 §7.5 step 4).
3. Write where it raced on the work order.
4. For the next part on this mold, tape a dam at that edge.

The mesh break is the built-in defence: flow mesh stops 30–50 mm short of the outlet side, which slows resin at the end and cuts race-tracking through the exit (CS-006 §7.2 step 3).

If resin fills the resin trap fast, it is over-pulling past full wet. The trap is doing its job: clamp the inlet earlier next time, and empty the trap before it nears the pump port (CS-006 §7.7).

7. Bubbles: foamy resin at the inlet, or bubbles under the bag

White, foamy resin at the inlet means the cup is slurping: re-submerge the line. Bubbles usually get trapped in the mesh, so note where (CS-006 §7.7). Bubbles under a bag that still holds: flush more resin through before clamping the inlet.

Also called: bubbles, air bubbles, foamy resin, froth, voids, pinholes, slurping inlet.

Symptom	Likely cause	Do (CS-006 §7.7)
White or foamy resin at the inlet	Bubbles from a slurping cup	Re-submerge the line; note where bubbles trap in the mesh
Bubbles under a sound bag	Air drawn in during the pull	If the bag still holds, flush more resin through before clamping the inlet (shop practice, Larry, #composites 2026-04-03)

1. Keep the inlet line end fully submerged in the cup for the whole infusion.
2. Brace the cup so it cannot tip.
3. If you see bubbles, write where on the work order, and whether the resin was degassed.

Pinholes from the same mold before? Degassing the resin is optional and covered in QR-002 (degassing).